Trading Energy Futures with Inflation

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1 Introduction

The following strategies are built using energy futures and various inflation measurements. Since CPI data is released at low frequency, the 5y5y forward inflation measurement works as a good proxy for modeling energy returns. With more advancements of this model more inflation-related factors have been integrated. The most recent is the Bloomberg Intelligence Inflation Surprise indices for the US and the UK.

security	Start Date	End Date
CL1	1990-01-03	2024-10-29
CO1	1990-01-03	2024-10-29
HO1	1990-01-03	2024-10-29
NG1	1990-04-05	2024-10-29
QS1	1990-01-03	2024-10-29
XB1	2005-10-05	2024-10-29

2 Background Analysis

Start by running a regression of the raw values (lagged) against the returns. In this case they values are differenced (and log applied to 5y5y forward). Since commodity returns can be quite volatile and vary cross-sectionally, volatility targeting is used. In this case, a 10% volatility target will suffice. Due to high intraday volatility swings, two methods of volatility targeting will be used. The first is lagged targeting, which is the true "real" returns, regarding the fidelity of the backtest. Perfect targeting is implemented as well since it captures the intraday volatility targeting. The futures data is data is daily returns. At times lagged volatility target leads to huge swings due to intraday volatility, although perfect targeting is not fully achievable.

Table 1: Raw Returns Regression Result

		Param Val				p-value				t-stat			
Fut	Exog param	UK 5y5y	UK Surp	US 5y5y	US Surp	UK 5y5y	UK Surp	US 5y5y	US Surp	UK 5y5y	UK Surp	US 5y5y	US Surp
CL1	α	0.000	0.000	0.000	0.000	0.419	0.223	0.535	0.142	0.809	1.220	0.620	1.467
	β	0.040	-0.003	0.008	-0.009	0.178	0.329	0.640	0.230	1.347	-0.976	0.468	-1.200
CO1	α	0.000	0.000	0.000	0.000	0.379	0.215	0.507	0.169	0.880	1.240	0.663	1.377
	β	0.033	-0.005	0.016	-0.010	0.195	0.091	0.260	0.140	1.297	-1.689	1.127	-1.475
HO1	α	0.001	0.001	0.000	0.001	0.098	0.065	0.203	0.021	1.655	1.845	1.273	2.301
	β	0.042	-0.006	0.010	-0.012	0.104	0.050	0.467	0.084	1.628	-1.959	0.728	-1.727
NG1	α	0.000	0.000	0.000	0.001	0.656	0.719	0.671	0.502	0.445	0.359	0.424	0.671
	β	-0.022	-0.005	0.011	-0.010	0.783	0.566	0.808	0.622	-0.275	-0.574	0.243	-0.492
QS1	α	0.000	0.000	0.000	0.000	0.287	0.179	0.484	0.129	1.065	1.344	0.699	1.520
	β	0.056	-0.006	0.011	-0.010	0.022	0.046	0.406	0.136	2.286	-1.996	0.831	-1.491
XB1	α	0.001	0.001	0.001	0.001	0.048	0.044	0.055	0.044	1.979	2.017	1.921	2.016
	β	0.022	-0.004	0.021	-0.003	0.453	0.175	0.194	0.743	0.751	-1.358	1.299	-0.328

Table 2: Lagged Volatility Targeting Regression Results

		Param Val				p-value				t-stat			
Fut	Exog param	UK 5y5y	UK Surp	US 5y5y	US Surp	UK 5y5y	UK Surp	US 5y5y	US Surp	UK 5y5y	UK Surp	US 5y5y	US Surp
CL1	α	0.000	0.000	0.000	0.000	0.713	0.400	0.870	0.259	0.368	0.841	0.164	1.129
	β	0.009	-0.001	0.004	-0.002	0.215	0.142	0.309	0.237	1.239	-1.467	1.017	-1.182
CO1	α	0.000	0.000	0.000	0.000	0.535	0.318	0.681	0.242	0.621	0.998	0.412	1.170
	β	0.010	-0.002	0.007	-0.002	0.160	0.058	0.100	0.269	1.406	-1.893	1.646	-1.106
HO1	α	0.000	0.000	0.000	0.000	0.173	0.107	0.305	0.043	1.361	1.612	1.027	2.025
	β	0.012	-0.002	0.005	-0.002	0.117	0.026	0.259	0.221	1.569	-2.232	1.128	-1.223
NG1	α	0.000	0.000	0.000	0.000	0.437	0.443	0.449	0.302	0.778	0.768	0.756	1.032
	β	-0.007	-0.000	0.004	-0.002	0.714	0.870	0.710	0.745	-0.366	-0.164	0.372	-0.326
QS1	α	0.000	0.000	0.000	0.000	0.517	0.306	0.760	0.204	0.647	1.023	0.306	1.270
	β	0.015	-0.002	0.007	-0.002	0.047	0.015	0.102	0.293	1.985	-2.426	1.635	-1.052
XB1	α	0.000	0.000	0.000	0.000	0.059	0.055	0.063	0.055	1.886	1.920	1.862	1.919
	β	0.005	-0.001	0.004	-0.001	0.526	0.129	0.301	0.762	0.634	-1.520	1.033	-0.302

Table 3: Perfect Volatility Targeting Regression Results

		Param Val				p-value				t-stat			
Fut	Exog param	UK 5y5y	UK Surp	US 5y5y	US Surp	UK 5y5y	UK Surp	US 5y5y	US Surp	UK 5y5y	UK Surp	US 5y5y	US Surp
CL1	α	0.000	0.000	0.000	0.000	0.617	0.311	0.772	0.188	0.500	1.014	0.290	1.317
	β	0.009	-0.001	0.004	-0.002	0.214	0.128	0.347	0.217	1.243	-1.521	0.940	-1.234
CO1	α	0.000	0.000	0.000	0.000	0.402	0.216	0.532	0.154	0.838	1.237	0.626	1.427
	β	0.010	-0.002	0.006	-0.002	0.152	0.050	0.121	0.256	1.432	-1.957	1.550	-1.135
HO1	α	0.000	0.000	0.000	0.000	0.073	0.040	0.149	0.013	1.792	2.055	1.443	2.489
	β	0.011	-0.002	0.005	-0.002	0.118	0.016	0.208	0.186	1.563	-2.410	1.259	-1.322
NG1	α	-0.000	-0.000	-0.000	0.000	0.535	0.544	0.496	0.927	-0.621	-0.607	-0.681	0.092
	β	0.001	-0.000	0.002	-0.002	0.853	0.621	0.522	0.309	0.186	-0.494	0.641	-1.017
QS1	α	0.000	0.000	0.000	0.000	0.300	0.152	0.500	0.091	1.036	1.432	0.674	1.692
	β	0.014	-0.002	0.007	-0.002	0.049	0.010	0.084	0.301	1.965	-2.563	1.729	-1.035
XB1	α	0.000	0.000	0.000	0.000	0.024	0.022	0.026	0.022	2.259	2.295	2.234	2.294
	β	0.004	-0.001	0.004	-0.001	0.546	0.110	0.253	0.753	0.604	-1.598	1.144	-0.315

3 Trading Z-Scores

The original setup of this strategy and research was to investigate US 5y5y forward inflation. UK 5y5y forwards and the Bloomberg Economics Inflation Surprise were collected as extra signals to trade. The rest of this paper will show results for all groups, but during the initial research process, US 5y5y z-score was the only signal of interest.

This section will cover trading a generic 30 day exponentially weighted z-score applied to each signal used for all of the energy futures. Then apply a 10% volatility target to each strategy using the same lagged and perfect method stated above.

Below is a table of sharpes for the individual portfolios trading the z-scores

	UK 5y5y		UK Sur. Inf.		US 5y5y		US Sur. Inf.	
Vol. Target security	Lagged	Perfect	Lagged	Perfect	Lagged	Perfect	Lagged	Perfect
CL1	0.377	0.357	0.062	0.049	0.705	0.716	-0.466	-0.480
CO1	0.471	0.472	0.124	0.111	0.674	0.687	-0.430	-0.439
HO1	0.343	0.430	-0.127	-0.110	0.523	0.637	-0.407	-0.391
NG1	0.368	0.304	-0.220	-0.114	0.250	0.126	0.047	-0.317
QS1	0.438	0.518	0.047	0.020	0.761	0.878	-0.353	-0.333
XB1	0.203	0.271	-0.137	-0.192	0.456	0.554	-0.409	-0.372

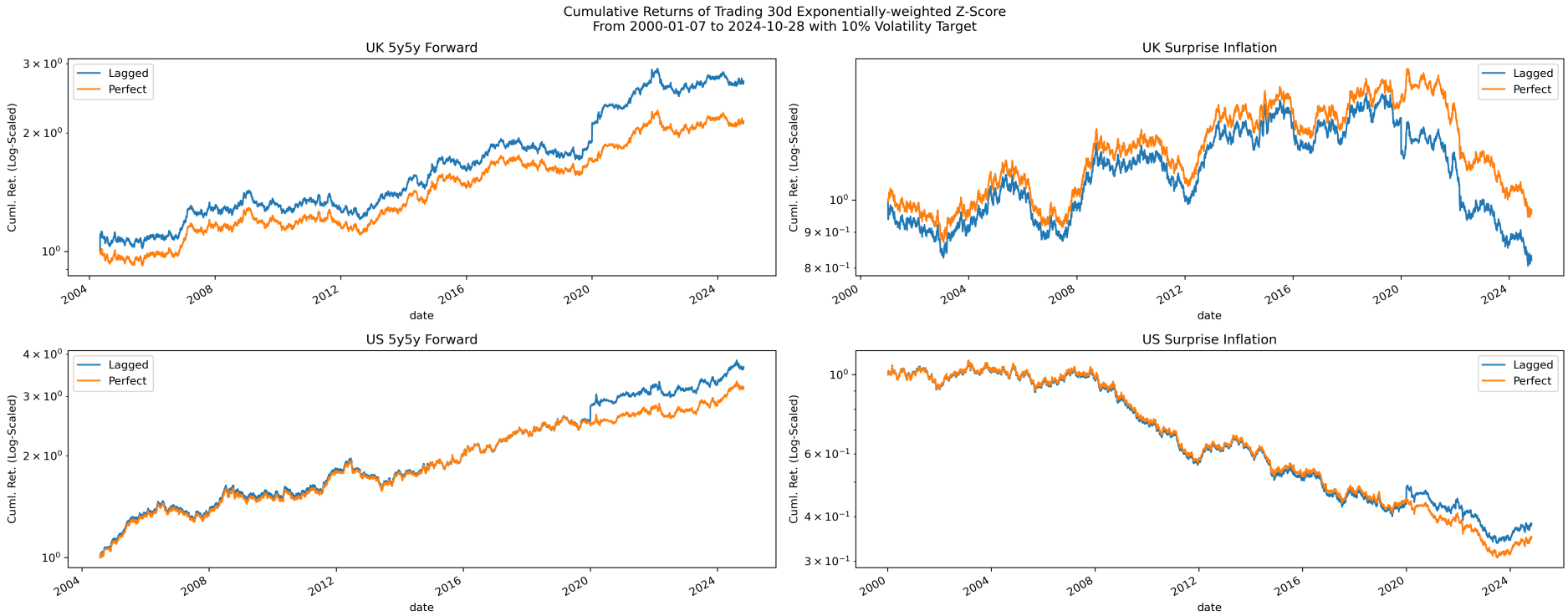
The results are not that strong. Now combine the strategies together per each factor and then calculate the sharpe per each group

Vol. Target	UK 5y5y	UK Sur. Inf.	US 5y5y	US Sur. Inf.
Lagged	0.331	-0.068	0.452	-0.241
Perfect	0.395	-0.032	0.605	-0.389

The results are being long the z-scores with the original idea that if the inflation measure richens (positive z-score) it is expected that the energy future will follow. The Bloomberg inflation surprise index measures how far away the analysts under- or over-estimate and therefore there is not an economically intuitive argument for trading the z-score one way or the other.

An argument can be made that analysts tend to overshoot being too pessimistic or optimistic and bet that they will regress back (i.e. shorting z-scores). On the other hand an argument can be made that when analyst increase or decrease their inflation expectation commodities will naturally follow.

Heuristically, an assumption about trading the z-scores of the inflation surprise cannot be made without fitting the signal or sampling the data. The following is a plot of the returns of the portfolios (per each factor).


Cumulative Returns of Trading 30d Exponentially-weighted Z-Score
From 2000-01-07 to 2024-10-28 with 10% Volatility Target
UK 5y5y Forward
UK Surprise Inflation
US 5y5y Forward
US Surprise Inflation
Lagged
Perfect
Cuml. Ret. (Log-Scaled)
date
3×10^0
2×10^0
10^0
4×10^0
9×10^{-1}
8×10^{-1}
6×10^{-1}
4×10^{-1}
3×10^{-1}
2000
2004
2008
2012
2016
2020
2024

4 Trading Optimized Z-Score

Now build an optimized trading strategy by calculating the deciles of the z-score and then calculating the sharpe per decile. Break the deciles into two groups an upper and lower consisting of the 1st/2nd and 9th/10th (respectively). If the upper and lower sharpes are the same sign, take a position according to the sharpe. Otherwise, no position.

The decile optimization works well in many ways, such as effectively residualizing the signal to avoid trading the noise, parametrically fitting the returns, de-correlating returns, and reducing transaction cost. Since the model is quite easy to calculate, the out-of-sample model is done expansively with a minimum observation window of 30 days.

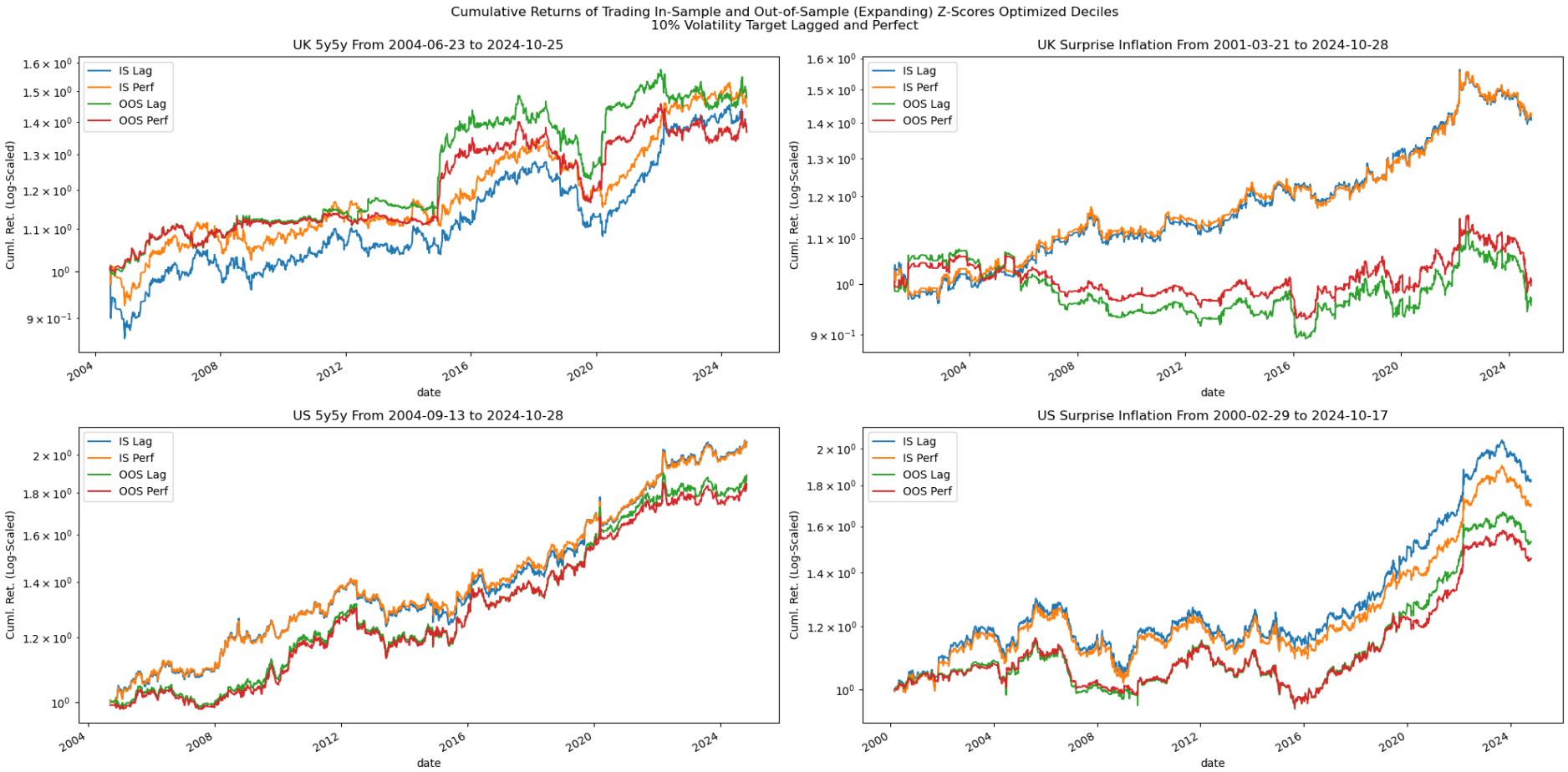
Below is a table of the portfolios' sharpe. The US 5y5y shows the best results of about 1.1-1.2 in-sample sharpe and 1-1.1 out of sample sharpe. That sharpe loss for 5y5y is consistent with real-life application of going from in-sample to out-of-sample. UK Surprise Inflation gives a cautionary tale of overfitting as the strategy loses almost of all of its sharpe in-sample.

	In-Sample		Out-Sample		Sharpe Lost	
Vol. Target Factor	Lagged	Perfect	Lagged	Perfect	Lagged	Perfect
UK 5y5y	0.410	0.441	0.577	0.536	0.167	0.095
UK Sur. Inf.	0.722	0.771	-0.040	0.009	-0.762	-0.762
US 5y5y	1.167	1.241	1.095	1.127	-0.072	-0.115
US Sur. Inf.	0.776	0.727	0.708	0.699	-0.068	-0.028

On the next page is a table of the sharpe of the individual signals controlling for each factor, lagged and perfect volatility targeting, and adjusted sharpe for times when not invested. On the following page is the returns of equal weight (of volatility targeted) returns.

Table 4: Sharpe of Returns of In-Sample and Expanding Out-of-Sample Optimized Z-Score S

			In-Sample				Out-Sample			
		Vol. Target	Lagged		Perfect			Lagged		Perfect
		Sharpe	Adjusted	Raw	Adjusted	Raw	Adjusted	Raw	Adjusted	Raw
security	group	country								
CL1	5y5y	UK	0.310	0.309	0.249	0.249	0.185	0.137	0.117	0.087
		US	1.082	1.080	1.074	1.072	1.007	0.952	0.998	0.943
	Sur. Inf.	UK	0.350	0.349	0.407	0.406	-0.015	-0.011	0.009	0.006
		US	0.801	0.566	0.704	0.498	0.575	0.446	0.641	0.497
CO1	5y5y	UK	0.512	0.363	0.489	0.347	-0.399	-0.303	-0.075	-0.057
		US	1.137	1.135	1.135	1.133	1.078	0.997	1.073	0.993
	Sur. Inf.	UK	0.247	0.174	0.320	0.226	-0.251	-0.181	-0.223	-0.160
		US	0.631	0.631	0.624	0.624	0.391	0.238	0.645	0.393
HO1	5y5y	UK	0.578	0.410	0.906	0.642	0.523	0.353	0.639	0.431
		US	1.510	1.065	1.523	1.074	1.026	0.967	1.014	0.956
	Sur. Inf.	UK	1.116	0.791	1.085	0.770	0.616	0.422	0.743	0.508
		US	0.518	0.518	0.477	0.477	0.294	0.214	0.705	0.514
NG1	5y5y	UK	-0.138	-0.138	0.269	0.268	0.394	0.383	0.453	0.441
		US	-0.048	-0.034	0.999	0.704	0.353	0.271	0.389	0.299
	Sur. Inf.	UK	0.398	0.397	0.463	0.462	0.097	0.090	0.218	0.201
		US	NaN	NaN	NaN	NaN	-0.144	-0.132	0.641	0.584
QS1	5y5y	UK	0.411	0.289	0.638	0.449	0.762	0.584	0.763	0.584
		US	1.208	1.199	1.280	1.270	1.404	1.318	1.455	1.365
	Sur. Inf.	UK	0.806	0.566	0.825	0.579	-0.149	-0.132	-0.213	-0.190
		US	0.638	0.634	0.575	0.571	0.442	0.342	0.346	0.267
XB1	5y5y	UK	0.446	0.446	0.519	0.519	0.271	0.256	0.247	0.233
		US	0.693	0.490	1.030	0.728	0.561	0.541	0.611	0.589
	Sur. Inf.	UK	1.300	0.918	1.357	0.958	0.074	0.054	0.163	0.120
		US	0.472	0.471	0.580	0.579	0.668	0.491	0.667	0.491


Cumulative Returns of Trading In-Sample and Out-of-Sample (Expanding) Z-Scores Optimized Deciles
10% Volatility Target Lagged and Perfect
UK 5y5y From 2004-06-23 to 2024-10-25
UK Surprise Inflation From 2001-03-21 to 2024-10-28
US 5y5y From 2004-09-13 to 2024-10-28
US Surprise Inflation From 2000-02-29 to 2024-10-17
IS Lag
IS Perf
OOS Lag
OOS Perf
Cuml. Ret. (Log-Scaled)
date

5 Trading OLS Residuals

The next area of advancement for this signal is to fit it to the returns via OLS and then trade the residuals. The fit of the OLS was not strong to begin with, from the above tables. The traded results were not very strong, so the signal was passed over.

Below is a table of the annualized sharpe of trading the raw residuals of the OLS. The resulting sharpes are not that strong.

Vol. Target	Lagged		Perfect	
	In-Sample	Out-of-Sample	In-Sample	Out-of-Sample
port				
UK 5y5y	-0.028	-0.031	0.133	0.143
UK Sur. Inf.	-0.043	0.006	0.148	0.264
US 5y5y	-0.112	-0.081	0.066	0.103
US Sur. Inf.	-0.071	-0.042	0.076	0.143

The OLS residuals were a good candidate for decile optimization, which was applied. The resulting portfolio was not very strong, the results were about 0.6-0.7 sharpe. The deciling is also done in-sample which would imply that out-of-sample the sharpe would be even lower. Below is a table of the returns deciled sharpe strategy.

	Vol. Target	Lagged		Perfect	
		Adj Sharpe	Raw Sharpe	Adj Sharpe	Raw Sharpe
factor	country				
5y5y	UK	0.391	0.244	0.539	0.373
	US	0.539	0.293	0.662	0.407
Sur. Inf.	UK	0.477	0.326	0.596	0.433
	US	0.510	0.352	0.636	0.474

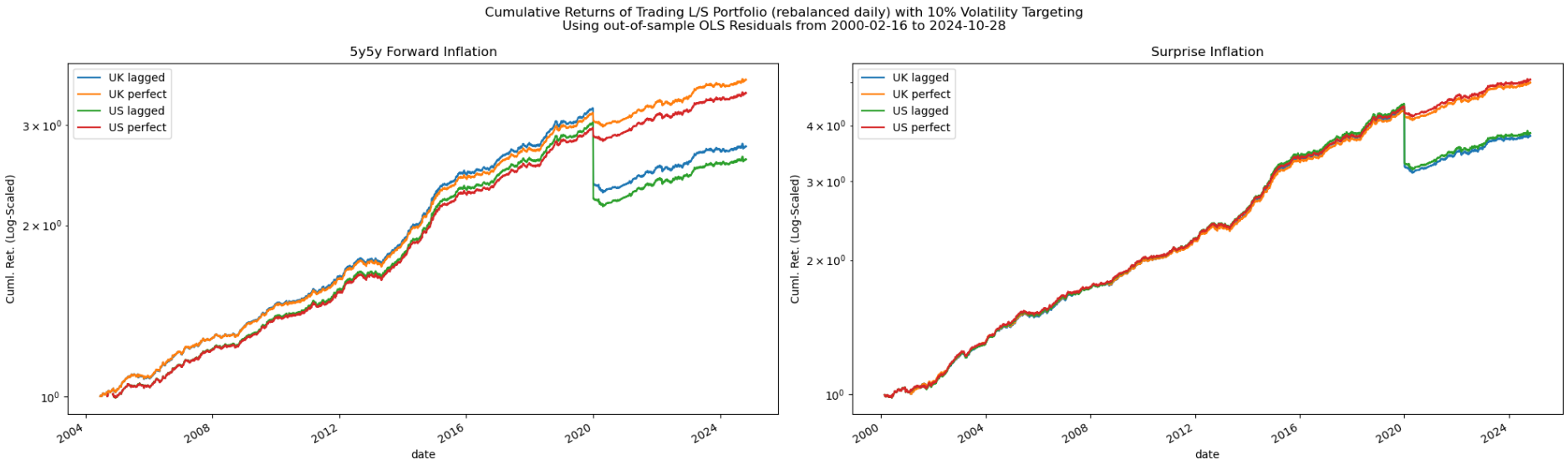
6 Cross Sectional Strategies

The next section of strategies to investigate is trading the signal cross-sectionally. Start by running the regression, getting the residuals, and sorting them into the portfolio. Due to lack of universe size within energy futures two groups will be used: upper residuals and lower residuals. Lagged and perfect volatility targeting are used. The prior method applied volatility targeting to signal returns this is applying volatility targeting to raw returns.

The resulting portfolio is quite strong. Using equal weight (equal volatility) of portfolio and rebalancing it daily leads to substantial returns. Below is a table of each leg and the long short portfolio of the in-sample residuals.

Vol. Target	Factor country	5y5y UK	5y5y US	Sur. Inf. UK	Sur. Inf. US
Lagged	Lower Resid	0.773	0.699	0.859	0.924
	Net Port	0.793	0.758	0.948	0.946
	Upper Resid	-0.241	-0.269	-0.297	-0.233
Perfect	Lower Resid	0.926	0.850	1.008	1.075
	Net Port	2.207	2.152	2.386	2.307
	Upper Resid	-0.527	-0.572	-0.565	-0.466

The results are quite strong for the perfect volatility targeting portfolio the net portfolio across all the factors has about 2 sharpe. The lagged volatility targeting portfolio has about 0.7-0.9 sharpe as well. This is due to an outsized loss that occurs during COVID. In real life, hedging would be done at the intraday level to avoid some of that loss. The next page plots the returns of the out-of-sample portfolios.


Cumulative Returns of Trading L/S Portfolio (rebalanced daily) with 10% Volatility Targeting
Using out-of-sample OLS Residuals from 2000-02-16 to 2024-10-28
5y5y Forward Inflation
Surprise Inflation
UK lagged
UK perfect
US lagged
US perfect
Cuml. Ret. (Log-Scaled)
date
10^0
2×10^0
3×10^0
4×10^0
2000
2004
2008
2012
2016
2020
2024